

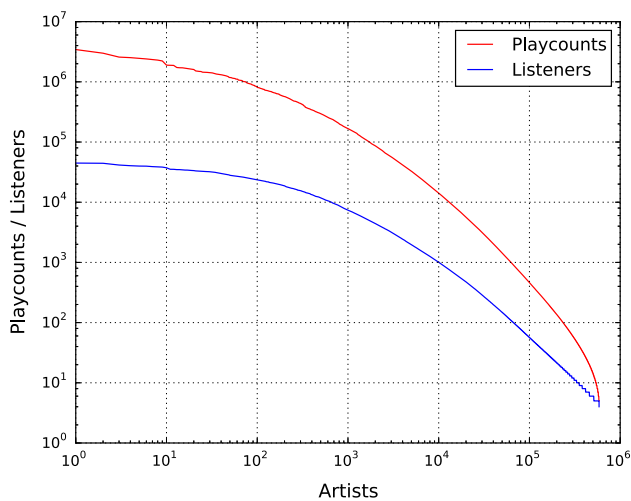


Fig. 2 Distribution of listening events by artist, log-log-scaled

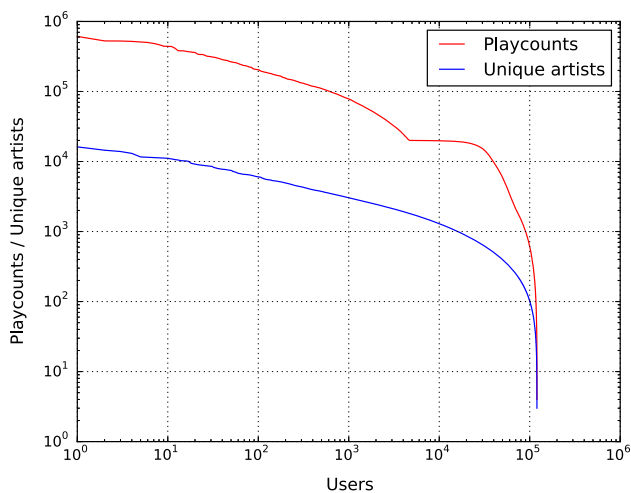


Fig. 3 Distribution of listening events by user, log-log-scaled

the median of the data by a horizontal red line. The lower and upper horizontal black lines of the box indicate the 25- and 75-percentiles, respectively. The horizontal black lines further above or below represent the furthest points not considered outliers, i.e., points within 1.5 times the interquartile range. Points beyond this range are depicted as blue plus signs. The red squares illustrate the arithmetic mean.

We can observe in Fig. 4 that the share of listening events does not substantially differ between working days. However, during weekend (Saturday and Sunday), there is a much larger spread. A majority of people listens less during weekends than during working days (lower median). At the same time, the top 25% of active listeners consume much more music during weekends (higher 75-percentile for Saturday, and even higher for Sunday). This is obviously the result of working and leisure habits.

In Fig. 5, we see that the distribution of listening events over hours of day vary more than over weekdays. It is partic-

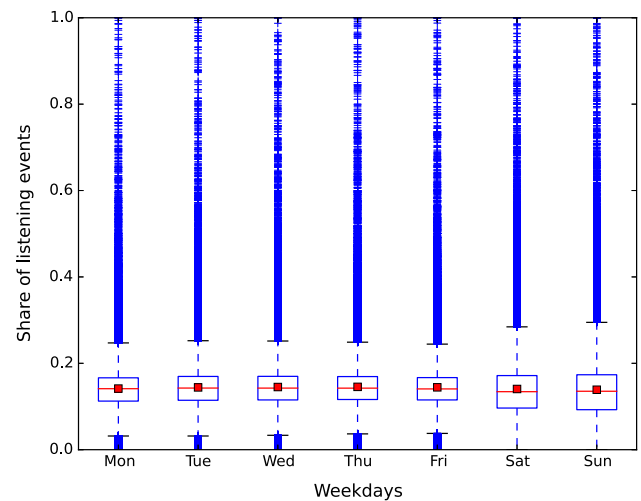


Fig. 4 Distribution of listening events over weekdays

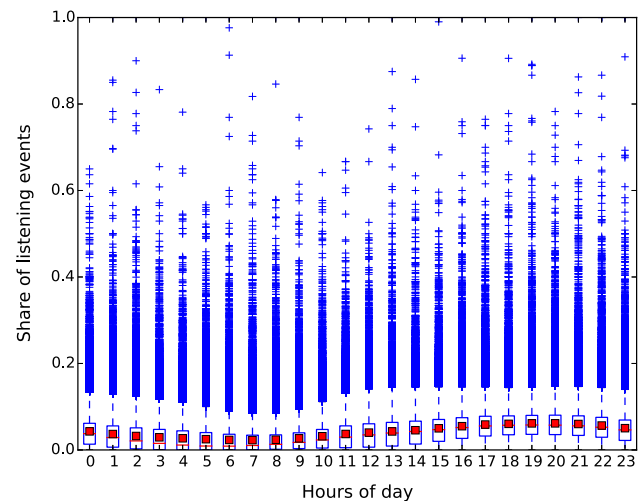


Fig. 5 Distribution of listening events over hours of day. Each time range encompasses 0–59 min after the hour indicated on the x-axis

ularly low during early morning hours (between 4 and 7h) and peaks in the afternoon and early evening (between 17 and 22h) when many people indulge in leisure time activities.

To compute the listener scores for novelty and mainstreamness, we follow the approach presented in [13]. For *novelty*, we split user u 's listening history into time windows of fixed length and calculate the percentage of new items listened to, i.e., items appearing for the first time in u 's listening history. The novelty N_{ut} of u 's listening events in time window t is defined as $N_{ut} = \frac{|l \in L_{ut} \wedge l \notin L_{ux} \forall x < t|}{|L_{ut}|}$, where L_{ut} is the entirety of items u listened to in time window t , including duplicates, and $l \notin L_{ux} \forall x < t$ denotes all listening events not listened to by u at any time before t . Averaging over all time windows user u was active in, we obtain u 's overall novelty score N_u . In the LFM-1b dataset, we provide novelty scores for time windows of 1, 6, and 12 months. To quantify